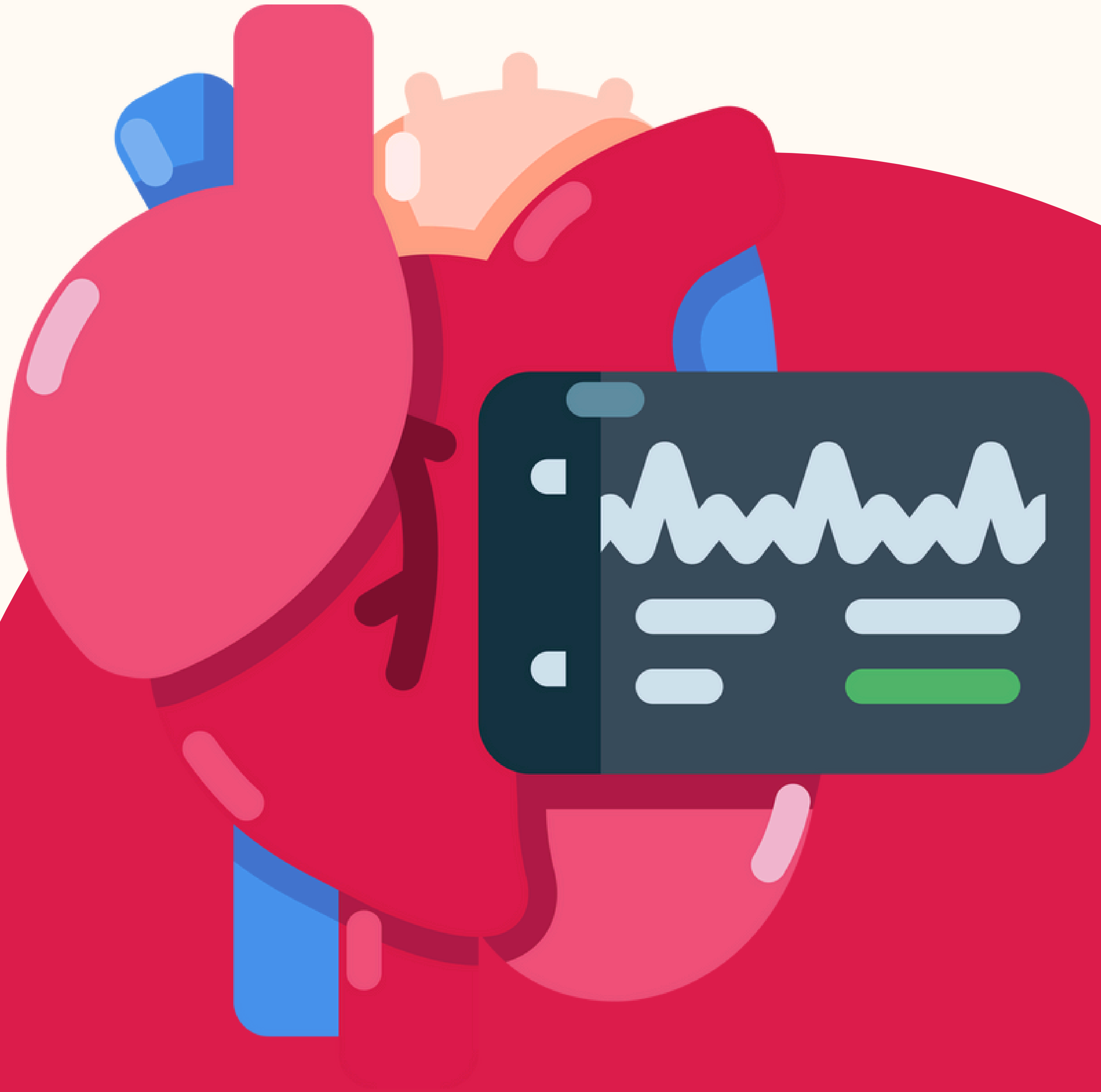
EARLY DETECTION OF HEART DISEASES WITH ECG SIGNALS USING MACHINE LEARNING



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PROJECT OVERVIEW

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GENERAL INTRODUCTION

According to the World Health Organization (WHO), in 2019, an estimated 17.9 million people died from cardiovascular diseases globally. This represents about 32% of all global deaths. Keep in mind that these figures can vary by region and country.

An electrocardiogram (ECG) is a signal that measures the electric activity of the heart. The proposed approach is implemented using ML-libraries and Scala language on Apache Spark framework; ML-lib is Apache Spark's scalable machine learning library. The key challenge in ECG classification is to handle the irregularities in the ECG signals which is very important to detect the patient status

APPLICATIONS

- Measurement of the electrical activity of the heart to detect cardiac problems.
- It can be helpful for the self-diagnosis
- To help the doctors with research studies

PROBLEM STATEMENT

Early detection of arrhythmias before major heart failures can reduce cardiac arrests in patients by suitable numbers.

The problem addressed in this project is the accurate prediction of heart status using electrocardiogram signals, which is essential for early diagnosis and treatment of cardiac diseases

LITERATURE SURVEY

Sl.No	Title	Authors	Conferences	Methods	Remarks
1	Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network	Awni Hannun, Pranav Rajpurkar, Codie Bourn, Mintu P. Turakhia & Andrew	Natural Medicine Journal	Convolutional Neural Network (CNN)	The paper has proven that there is potential for developing an end-to-end deep learning approach can classify a broad range of arrythmias.
2	Heartbeat classifcation system based on neural networks and dimensionality	Dalvi RdF, Zago GT, Andreão RV	Research on Biomedical Engineering-2016-	Neural network and SVM were used for classification.	Performance on a test set of only 18 ECG records of 30 min each achieved an accuracy of 96.97%.
3	A generic and robust system for automated patient-specifc classifcation of ECG signals.	Ince T, Kiranyaz S, Gabbouj M.	IEEE Trans Biomed Eng-2009	Feed-forward and fully connected artifcial neural networks aided by particle swarm optimization.	Two patterns of heatbeats are recognised with 96% accuracy.

LITERATURE SURVEY

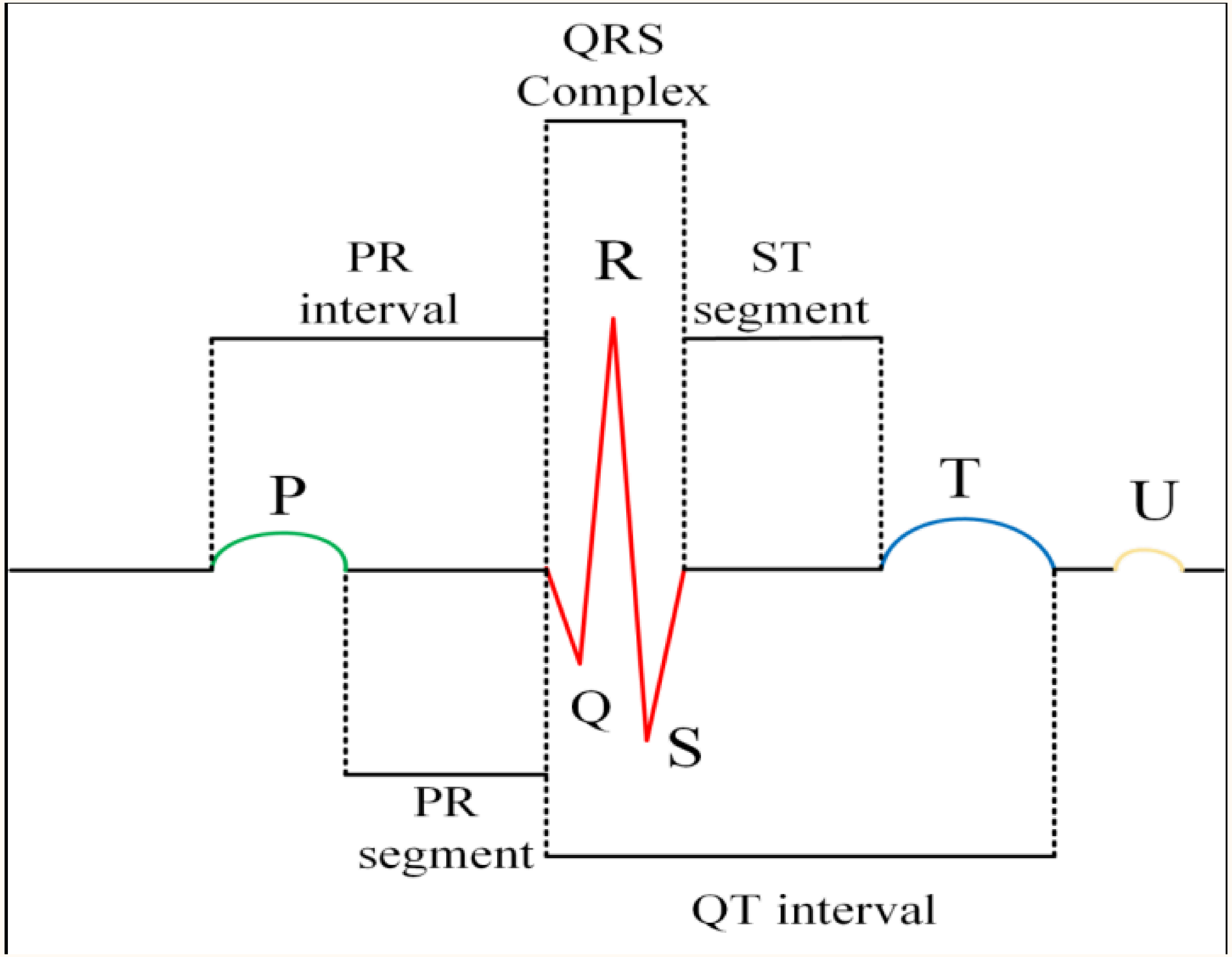
Sl.No	Title	Authors	Conferences	Methods	Remarks
4	Big data analytics in genomics: the point on deep learning solutions.	Celesti F, Celesti A, Carnevale L, Galletta A, Campo S, Romano A, Bramanti P, Villari M. B	2017 IEEE symposium on computers and communications (ISCC)	DNN (Deep Neural Network) has been used for deep learning to classify heartbeats.	Classifcation was only two types (Normal and Abnormal) and dataset size was almost 85,000 records.
5	ECG heartbeat classifcation: a deep transferable representation	Kachuee M, Fazeli S, Sarrafzadeh M.	2018 IEEE international conference on healthcare informatics	Convolutional neural network for classifcation of ECG beat types has been developed by the author.	Multiply types of heartbeats have been studied and the author has reached accuracy 93.4%.
6	Using hidden Markov models and spark to mine ECG data	O'Brien J.	Saint Mary's College of Maryland	Combining accurate Hidden Markov models (HMM) techniques with Apache Spark to improve the speed of ECG analysis.	The paper has proven that there is potential for developing a fast classifer for heartbeats classifcation

WHAT IS ECG?

- An electrocardiogram (ECG) is a complete representation of the electrical activity of the heart on the surface of the human body, and it is extensively applied in the clinical diagnosis of heart diseases.

- ECG signals have been widely used for detecting heart diseases due to it's simplicity and non-invasive nature.

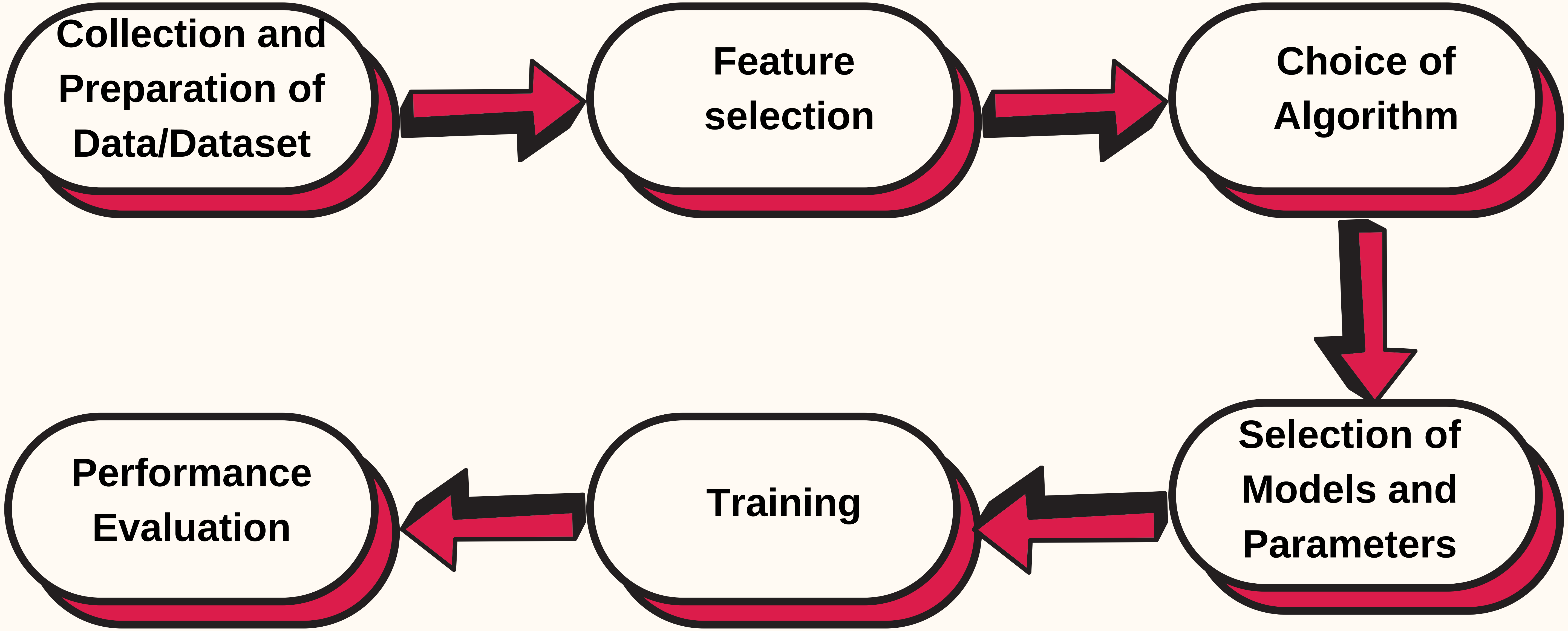
Components of ECG



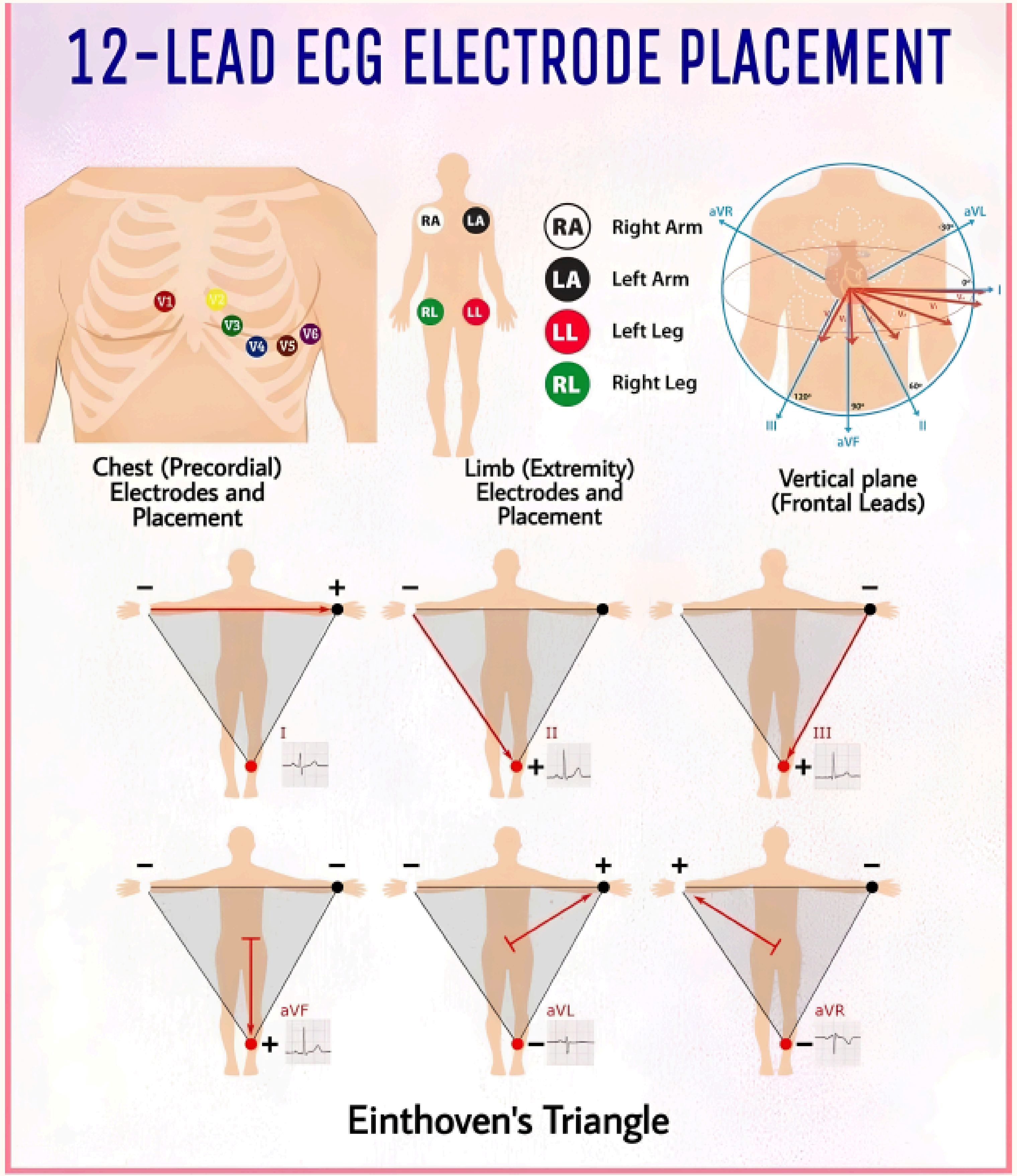
DESIGN OBJECTIVE

- To develop a machine learning model using convolutional neural network (CNN) that achieves high accuracy in classifying a broad range of distinct arrhythmias.
- The CNN should demonstrate the ability to generalize its learning from the training dataset to new and unseen ECG data.
- The design should prioritize computational efficiency to enable real-time or near-real-time analysis of ECG data.

GENERIC MODEL OF ML



GENERATION OF ECG



The Einthoven Triangle is a concept in electrocardiography (ECG) that refers to the spatial relationship between the three standard limb leads: Lead I, Lead II, and Lead III. These leads are used to record the electrical activity of the heart.

Lead I is positioned between the right arm and left arm, Lead II is between the right arm and left leg, and Lead III is between the left arm and left leg. The Einthoven Triangle is an imaginary equilateral triangle formed by connecting the positive electrode of Lead I, the positive electrode of Lead II, and the positive electrode of Lead III.

Arrythmia: Heart disease caused by irregular heartbeat

Generally arrythmia refers to irregular heart beat it also refers to improper functioning of the heart beating of too fast or too slow.

Now see lets see the classification of averythmias for better understanding

• They are basically three types of Arrythmia

1)Sinus rhythms - is Classified into two types
i) Bradycardia : Generally Bradycardia is characterized by a slower than normal heartrate.
If you have bradycardia your heart beat is less than 60 times a minute.

ii) Tachycardia: In the Tachycardia a heart rate over 100 beats per minutes.

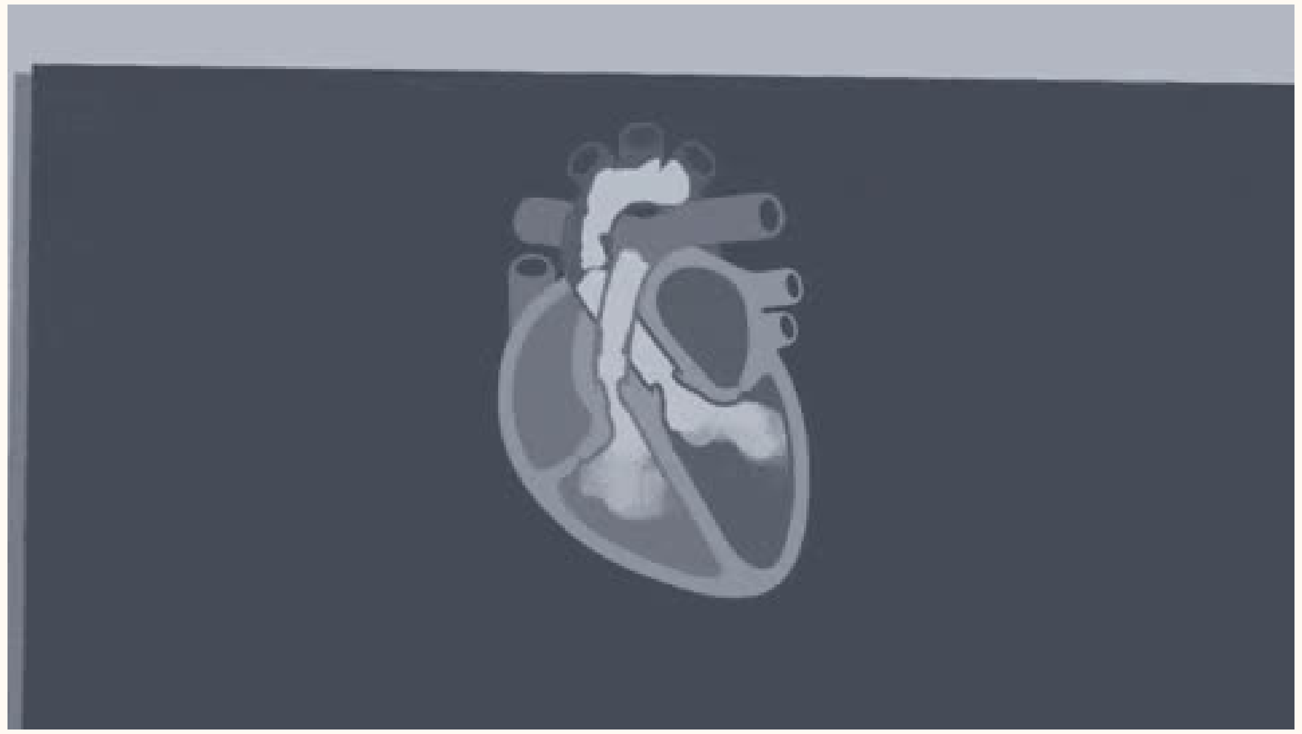
2.Atrial rhytms (it orginate from atria)

It is Classified into two i) Atrial fibrillation (AF) : This is the most common type, characterized by rapid, irregular benting of the upper chambers (atria).

AF Can lead to blood clots and increase the risk of stroke.

ii) Atarial flutter: It is similar to Af, but with more organized rythm.

It can cause palpitations and lead to complication like stroke it left Untreated.



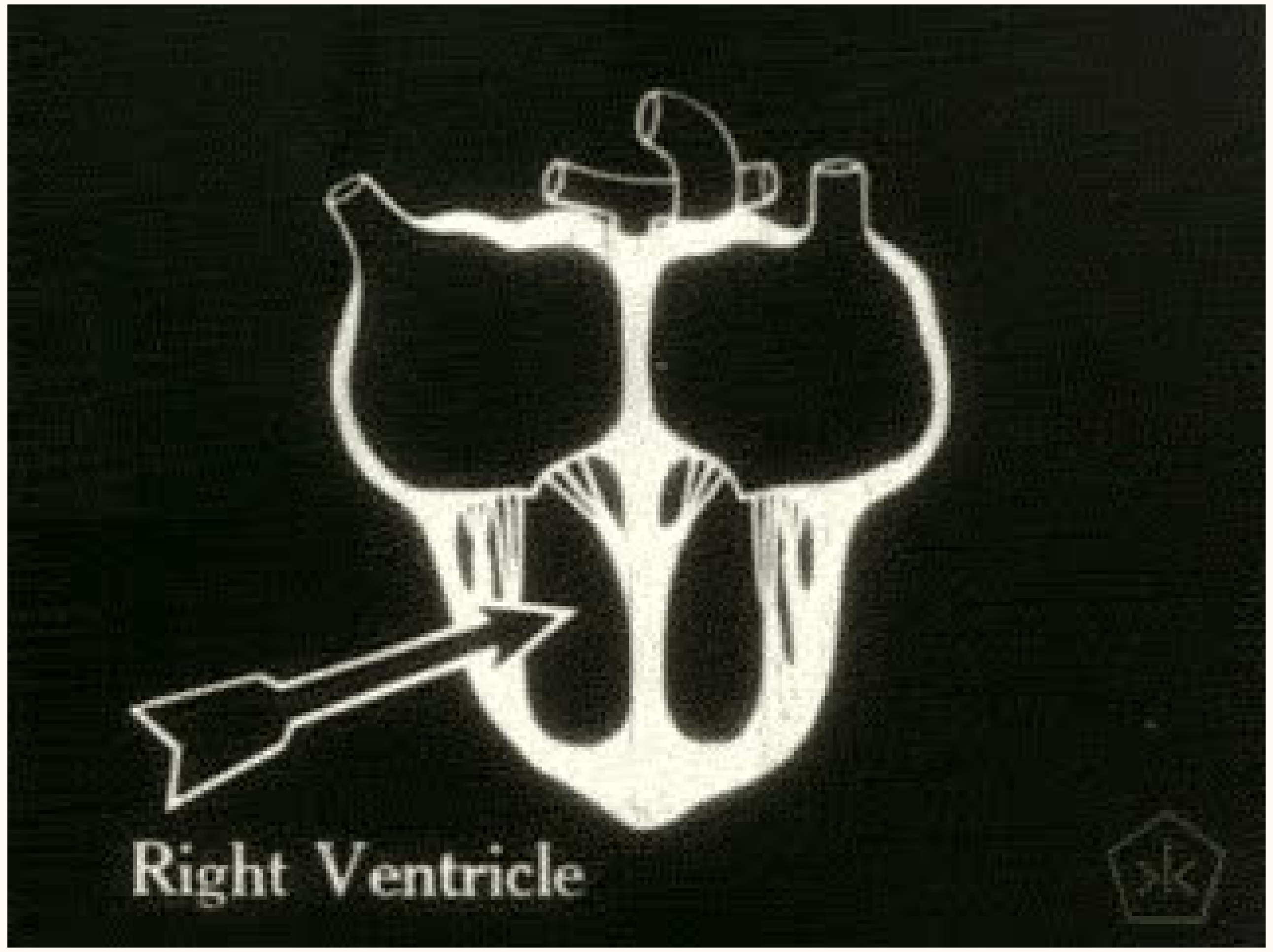
3.Ventricular rhythms (originate from the ventricles)

It is classified into two types

i) Ventricular tachycardia:-

It occurs due to a problem in the hearts electrical impulses.

- The condition may develop as a Complication of an heart attack
- In this disease, the lower chamber of the heart beat very quickly



Ventricular fibrillation (UF):

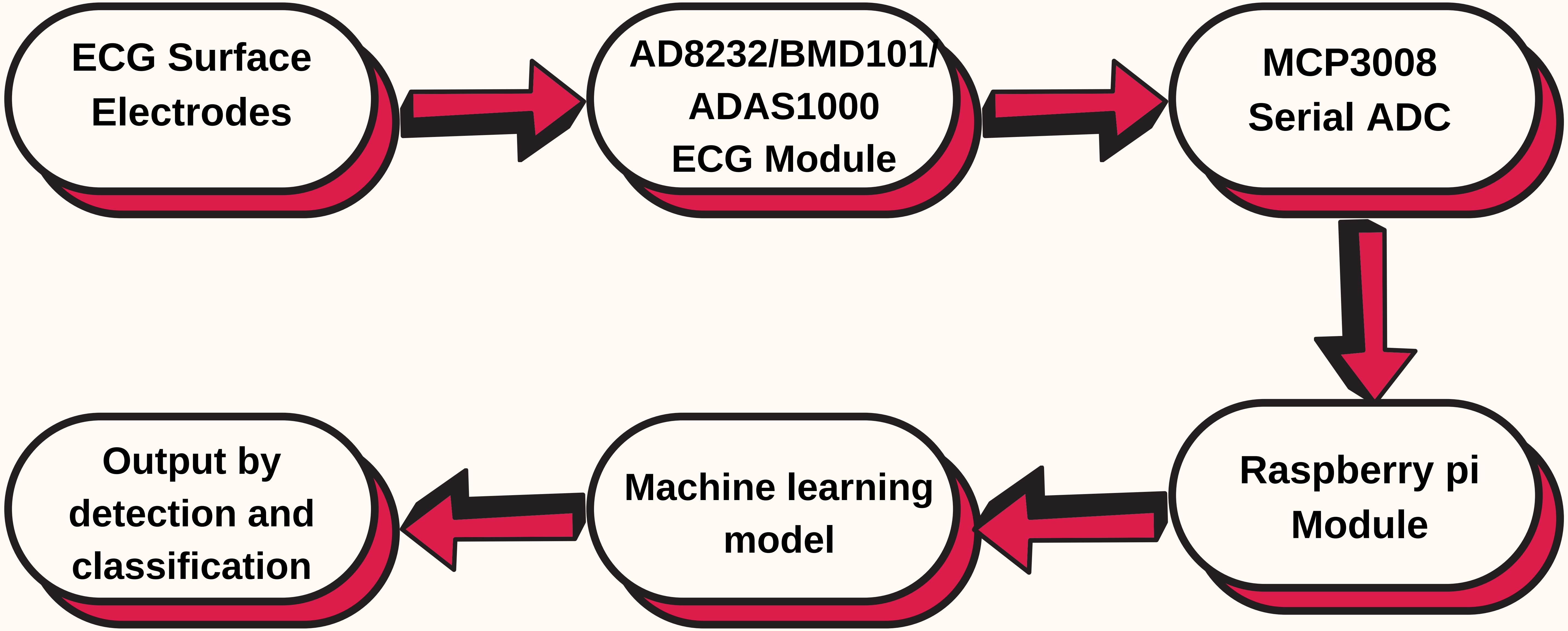
A life-threatening heart rhythm that results in a rapid, inadequate heartbeat.

VF is a rapid, life-threatening heart rhythm starting from lower chambers in the heart from this, a heart attack can be triggered.

DESIGN METHODOLGY

- Convolutional Neural Network are effective in capturing spatial and temporal in ECG signals
- The irregularities in ECG signals classify the heartbeats under different categories.
- The performance of ECG pattern depends upon the features and design of CNN model.
- The computational part is done on Raspberry pi using python libraries.

DESIGN SPECIFICATION AND HARDWARE COMPONENTS



MACHINE LEARNING

Machine learning is a subfield of artificial intelligence, which is broadly defined as the capability of a machine to imitate intelligent human behaviour.

- Improves accuracy.
- Computational Statistics to focus on making predictions.
  Mathematical Optimization which relates to:-
- Models, applications
- Frameworks to the field of statistics.
  Real-world problems make suitable applications.
- Speech Recognition
- Robotics and Artificial Intelligence
- Web and Social Media
- Healthcare

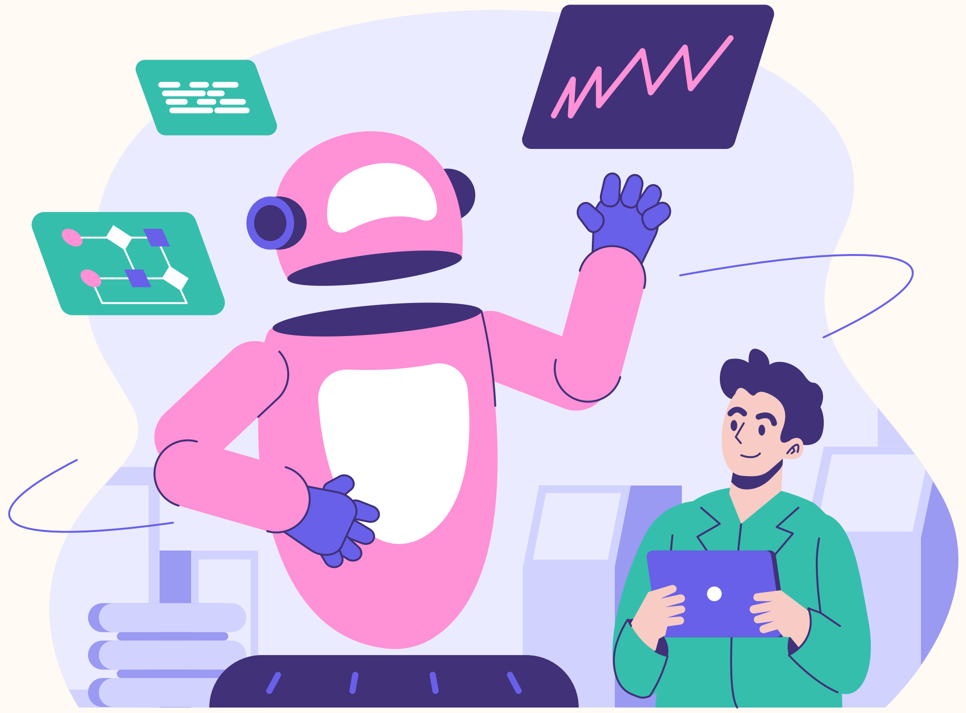
DEEP LEARNING- A SUBFIELD OF ML

Deep Learning

- A subfield of machine learning that focuses on training and building artificial neural networks with multiple layers to learn and represent complex patterns and relationships in data, enabling them to make accurate predictions or classifications.

The different approaches to ML

- <u>Neural Networks (NN)</u>
- Computational models inspired by the structure and function of biological brains.
- They consist of interconnected nodes, called neurons, organized into layers
- Each neuron receives input signals, performs computations, and generates an output signal



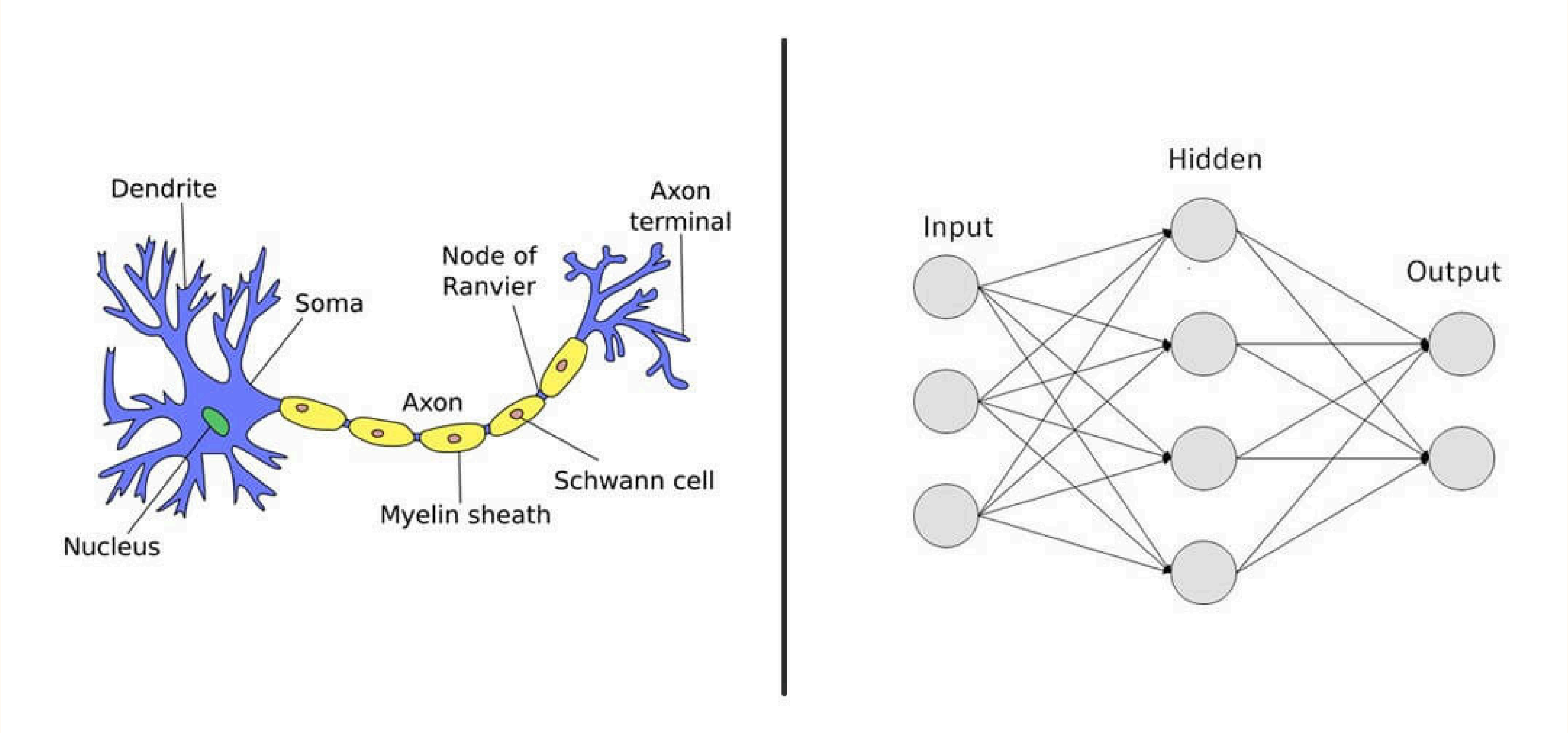
2. Deep Neural Network (DNN)

- Neural networks with multiple hidden layers.
- The depth of these networks allows them to capture indicate and abstract features in the data, leading to improved performance on complex tasks.

Types

i) Convolutional Neural Networks (CNN)

Specialized deep learning models commonly used for image and signal processing tasks.



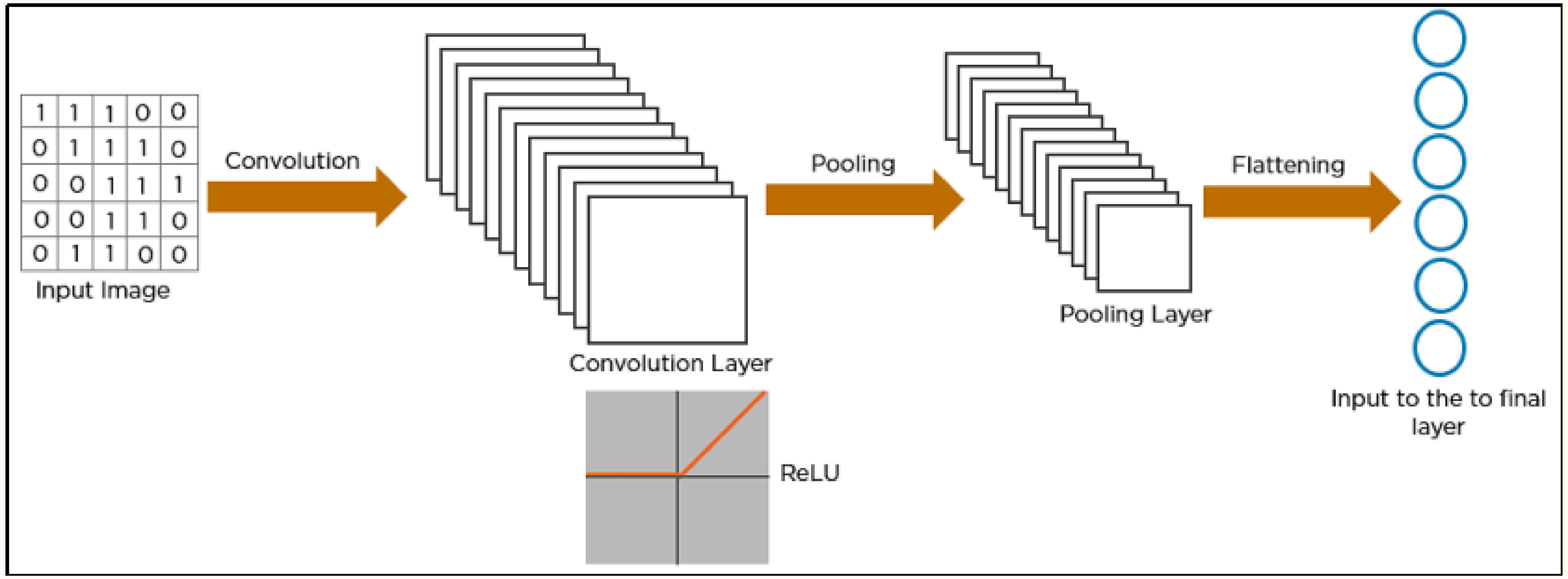
ii) Recurrent Neural Networks (RNN)

Deep learning models suitable for sequence data, such as time-series or natural language data.

Convolutional Neural Network (CNN)

- Convolutional Neural Networks (CNNs) are a type of deep learning architecture specifically designed for processing grid-like data, such as images or signals.
- CNNs have been widely used in computer vision tasks, including image classification, object detection, and image segmentation.
- They have also proven to be highly effective in various applications involving sequential data, such as natural language processing and time-series analysis, including electrocardiogram (ECG) analysis.

Architectuof Convolutional Neural Network



They have three main types of layers, which are:

1. Convolutional layer
2. Pooling layer
3. Fully-connected (FC) layer

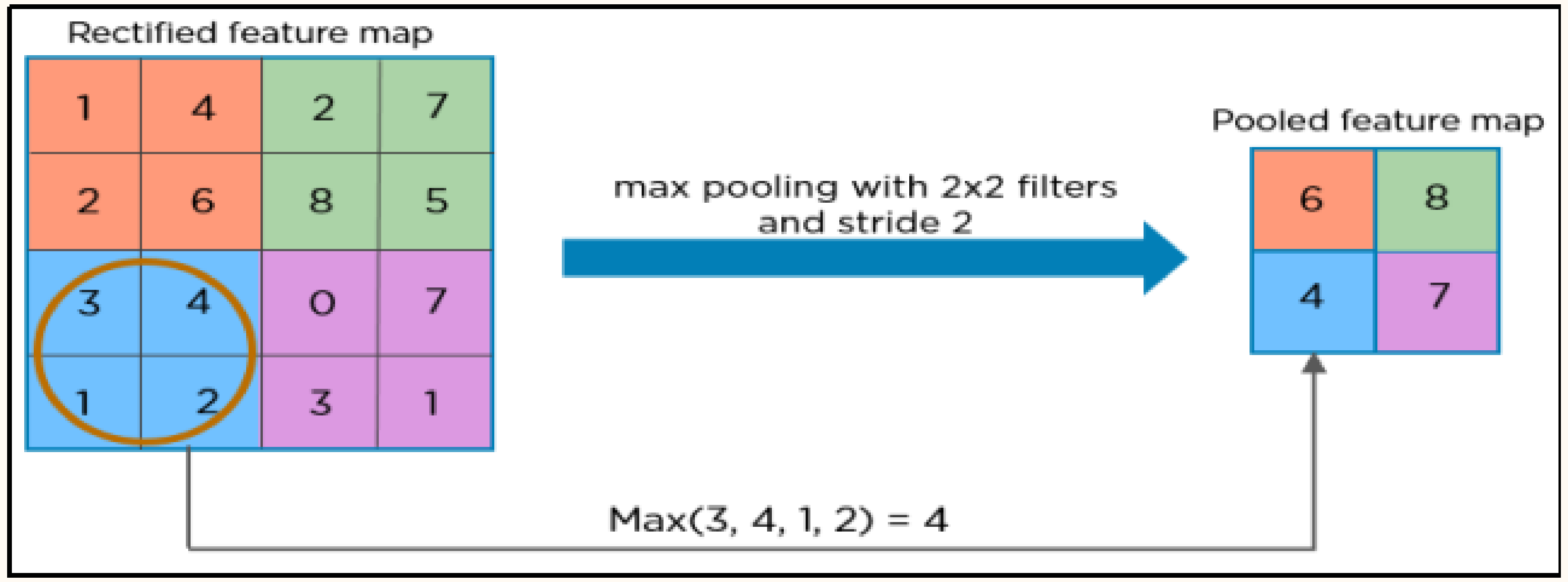
Working of Convolutional Neural Network

1.Convolutional Layer: - The core building block of a CNN is the convolutional layer.
- It applies a set of learnable filters to input data, enabling the network to automatically extract relevant features from ECG signals.

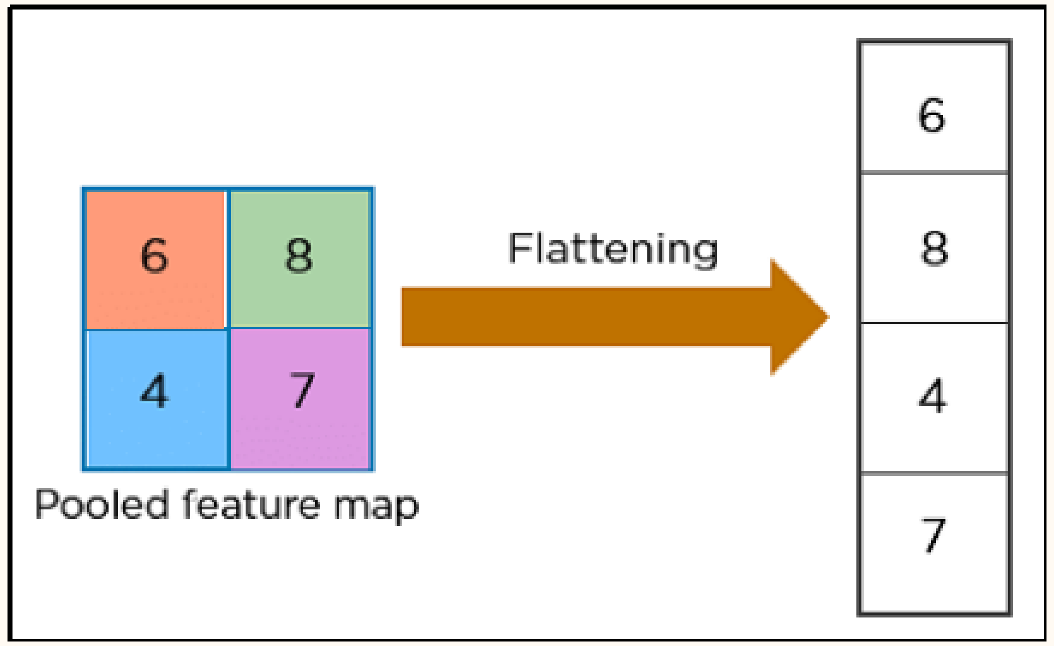
2.Pooling Layer: - Pooling layers are typically used after convolutional layers to reduce the spatial dimensionality of the feature maps while retaining the most important information.
-Max pooling, for example, selects the maximum value within a small neighborhood, effectively downsampling the feature maps.

3.Fully-Connected Layer: -These are typically employed to transform the high-level features learned from the earlier layers into class probabilities or regression predictions.
-They connect every neuron to every neuron in the previous and subsequent layers.

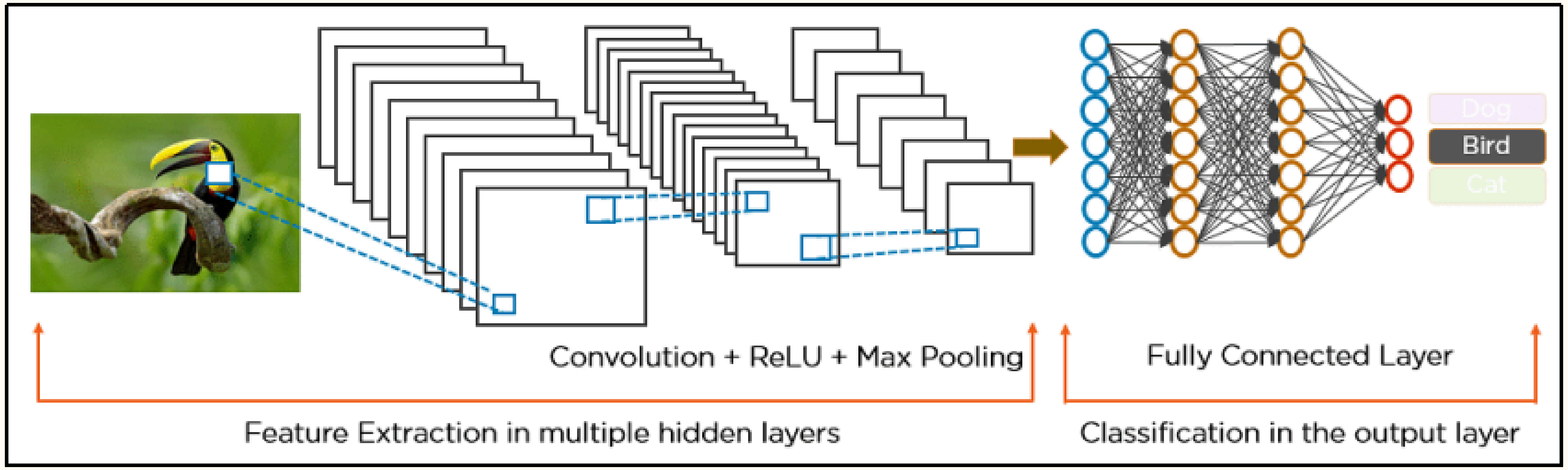
Pooling



Flattening



Example implementation of CNN

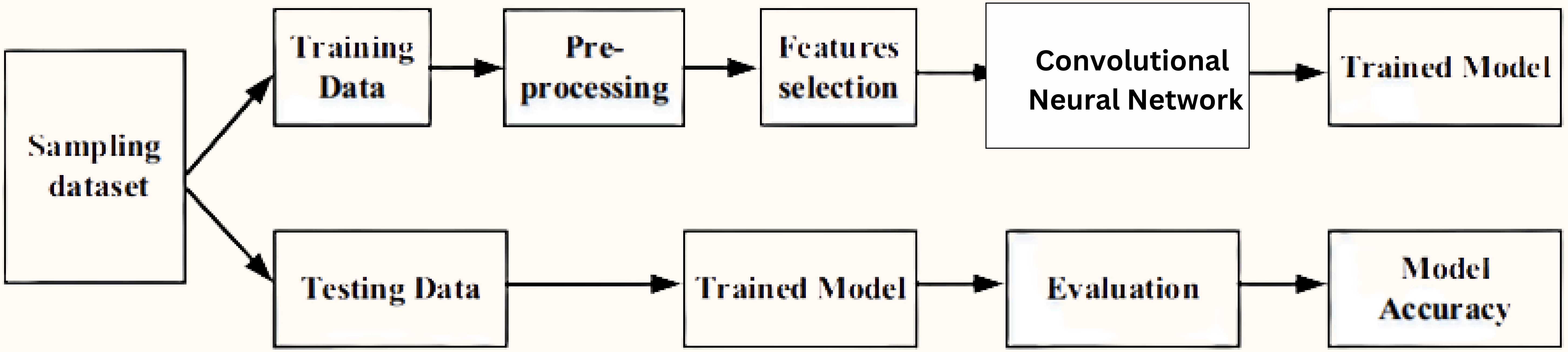


TOOLS

The proposed approach is implemented using ML-libs ,Tensor Flow,Keras and Python language on Flask framework.

- **TensorFlow** is a popular open-source deep learning library that provides comprehensive support for building and training neural networks, including CNNs. It offers high-level APIs, such as Keras.
- **Keras** is a high-level deep learning library that runs on top of TensorFlow or other backends.It offers a user-friendly API for designing and training CNN models with ECG signals.
- **Flask** is a micro web framework. It is used for a wide range of web applications, from simple static websites to complex web applications with dynamic content and user authentication.

DETAILED EXPLANATION



Libraries used

- Numpy
- Pandas
- Sklearn
- Matplotlib
- Scikit-Learn

Dataset Description

The dataset used consists of the following characteristics:

1. Size: The dataset contains a total of 91,232 single-lead electrocardiograms (ECGs) from 53,549 patients who underwent ambulatory ECG monitoring. Each ECG represents a unique patient.

2. Rhythm Classes: The dataset includes 12 different rhythm classes that were used for classification purposes.

Data Preprocessing

- The preprocess function defines the list of ECG record numbers and features, and if specified, splits the dataset into test and train sets based on the split parameter.

- Overall, the script downloads ECG data from the MIT-BIH Arrhythmia Database (if specified), preprocesses the data and extracts windows around peaks, retrieves annotations and saves the processed data and labels to HDF5 files. The script can be run directly to perform the preprocessing or imported as a module for reuse.

Mathematical equation’s

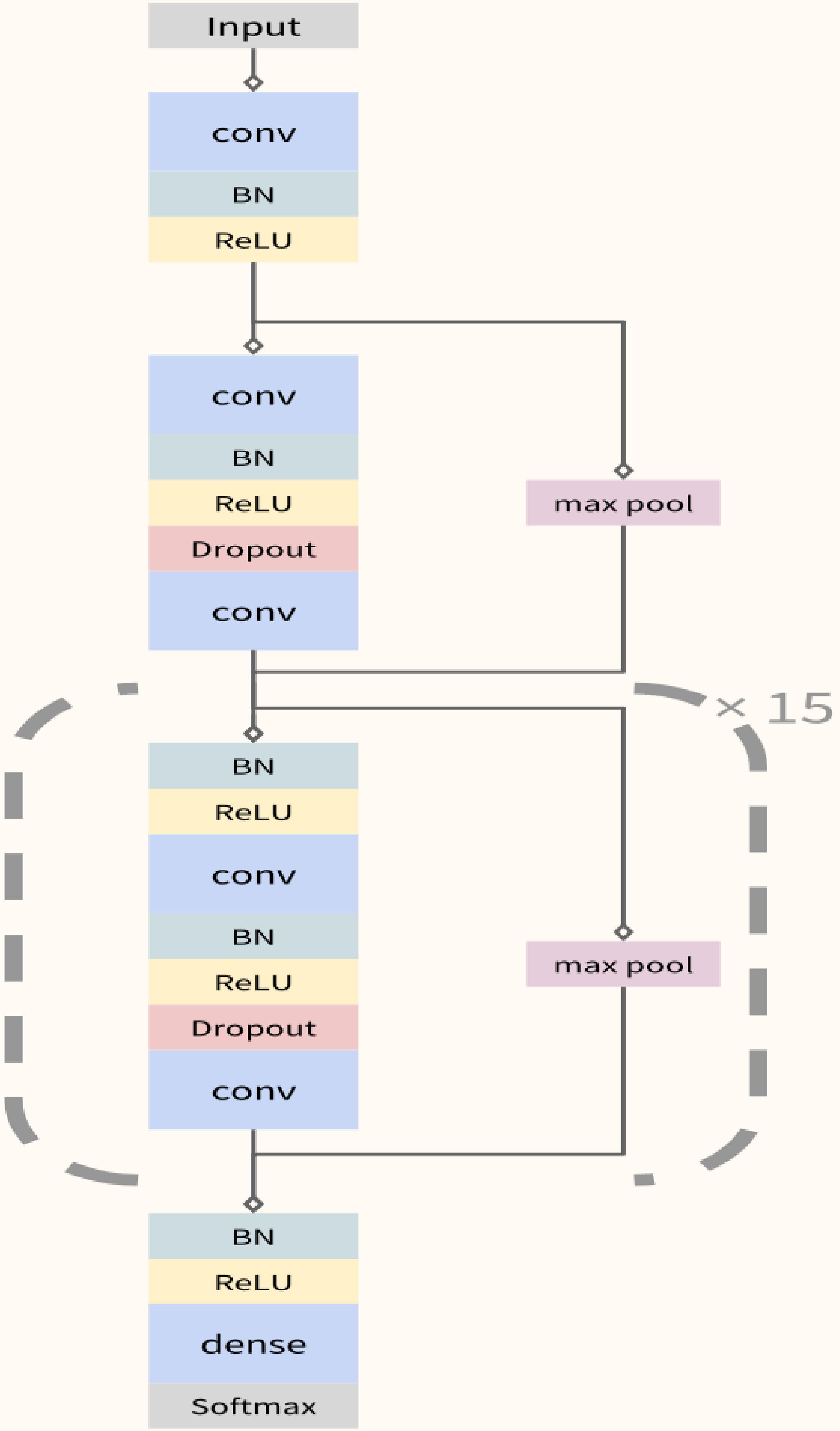
References	Monitoring Context	Evaluation method and/or Dataset	Evaluation parameter for Dataset				Study Criteria	Component validated or evaluated	Mathematical Formula	Results
			Data set name	No. of samples	Trained data samples	Dataset types				
[4]	Heartbeat segmentation	SVM based classifier	MIT-BIH, FECG, NSR,	1.5 M	500 K	ECG recording data type	Classification Accuracy	Activity Classification	$\varepsilon = \frac{1}{2}[(\nabla_u)^T + \nabla_u]$ ∇_u= Input features of the i_{th} training dataset of the class	96.96%
[3]	Heartbeat segmentation	RFE PTB ECG	ECG, HRV dataset	450M	35K	ANN dataset	Specificity	Dimensionality reduction	$\Delta_R(C_1, C_2) = \left[\sum_{i=1}^{n} log^2 \lambda_i\right]^{\frac{1}{2}}$ Where $n = Eigen - values\ of\ the\ matrix$ $C_1, C_2 = Martices$	86.58%.
[1]	Computer-aided diagnosis	MIT-BIH	PAMAP2, MIT-BIH	36M	1900	Sensor dataset	Efficiency	Evaluation	Classification algorithm $B_i = [\overline{L_i}, \overline{RSSI_i}, S_k](1 \le i \le N)$ $\overline{H} = Location\ ;\ \overline{RSSI_i} = RSSI\ profile;$ $S_k = Corresponding\ route$	78.66%
[7]	Soft Computing	DWT KDD CUP99	MIT-BIH, FECG, NSR, AAR	960K	360	ECG recording data type	Noise removal	Feature extraction	Paillier algorithm $Enc(m_1) + Enc(m_2) = Enc(m_1 + m_2)$ Enc = encryption operation $m_1 m_2$ = are two plaintext messages	76.03%
[9]	Cardiovascular disease classification	CNN NSL-KDDCUP	QT database, NSL	11,760	47,040	Encoder data type	Accuracy	Activity classification	Classifier $\varepsilon = \frac{1}{2}[(\nabla_u)^T + \nabla_u]$ ∇_u= Input features of the i_{th} training dataset of the class T = time	93.54%
[12]	Heartbeat classification	ECG Data	MIT-BIH, FECG, DS1, DS2	44060	2260	Recording Dataset type	Accuracy, sensitivity and specificity	Classification	Classification algorithm $CT_2 = M_2^d mod N_2$	95%, 100% & 80%

Mathematical equation’s

		AAR, NSR	AAR						T = time	
[5]	Human activity detection	FECG, MIT-BIH	PAMAP2, MIT-BIH	360M	19K	Sensor dataset	Accelerometer accuracy	Data pre-processing	$W_i = \frac{\sum_{j=1}^{n} a_{ij}}{n} and \sum_{j=1}^{n} W_i = 1$ i = Decision factor $a_{ij} = \frac{x_{ij}}{\sum_{i=1}^{n} x_{ij}}$	99%
[6]	Clinical decision-support	SVM	ECG, HRV dataset	450M	35K	ANN dataset	Prediction	Classification	$100 \times (2 \times max(p_0, p_1) - 1)$ p_0 – Probability of no heart disease p_1- Probability of heart disease	97%
[11]	Automated heartbeat classification	CNN	MIT-BIH, FECG, DS1, DS2	4400	202	Recording Dataset type	Accuracy	Data fusion	$EH = \sum_{k=1}^{m} (t_k - y_k)^2$ t_k= output of monitoring machine	99.10%
[2]	Cardiac Arrhythmias classification	K-Nearest Neighbour algorithm, CPSC, NSR	CPSC, NSR	963	896	Heart dataset	FI-Score	Evaluation	$Std = \sqrt{\frac{\sum_{i=1}^{n}(Diff_i - Mean)^2}{n}}$ x_i = extracted integer value , x^I = real integer value	80.95%
[8]	ECG signal delineation	SVM QT database, NSL	QT database, NSL	11,760	47,040	Encoder data type	Accuracy	Activity Classification	Classifier $\varepsilon = \frac{1}{2}[(\nabla_u)^T + \nabla_u]$ ∇_u= Input features of the i_{th} training dataset of the class T = time	90%
[10]	Arrhythmia analysis	HMM RBBB, APC	RBBB, ECG, MIT-BIH	200M	108K	APC arrhythmia dataset	Accuracy	Activity Classification	Classification $c_k = \sum_{j=1}^{n} [P_1 P_2 \dots P_k]_{ij} c_{k,j}$ P = Probability of transfer files n = 1,2,3,…, n k = 2n-bit string	99.8%

Model

- The model is mainly made of 3 blocks, the input plus a reset block, and a reset loop-block and a output block
- The network takes as input a time-series of raw ECG signal, and outputs a sequence of label predictions.
- The 30 second long ECG signal is sampled at 200Hz, and the model outputs a new prediction once every second.
- We arrive at an architecture which is 33 layers of convolution followed by a fully connected layer and a softmax.



Training and Testing

- A computer model is trained to analyze ECG data and classify different types of heartbeats.
- It takes into account the configuration, splits the data into training and validation sets (if needed), sets up the model and its training parameters, trains the model using the data, and evaluates its performance..

Evaluation of trained model

- The performance of the model is evaluated to assess how well it can classify different types of heartbeats using ECG data.
- After training the model, it is tested using validation data.
- The model predicts the type of each heartbeat, and these predictions are compared to the actual labels.
- Performance metrics like accuracy, precision, recall, and F1-score are calculated to measure how accurately the model classifies the heartbeats.

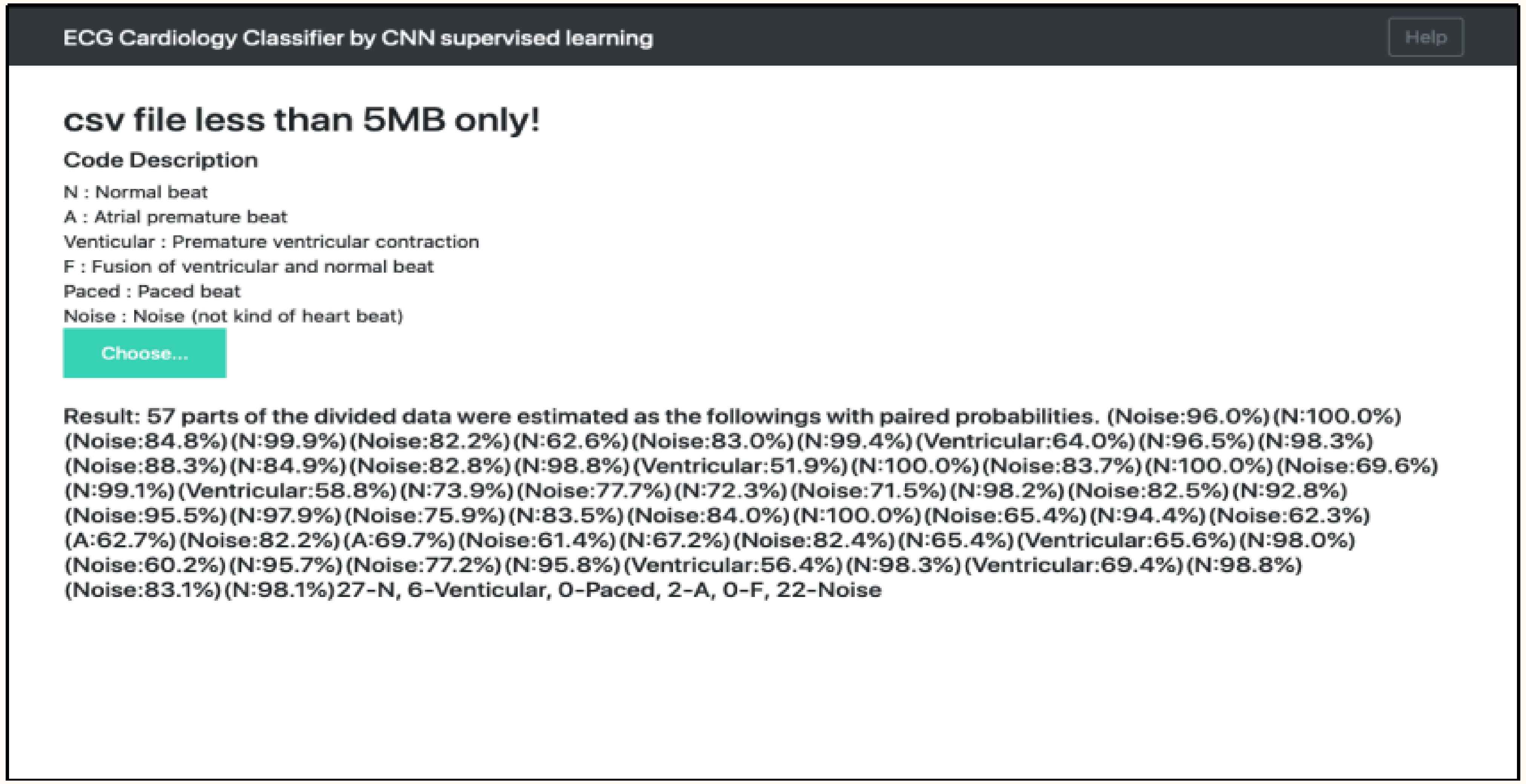
Predictions

Using the trained model, we can predict and compare the result with its true labels

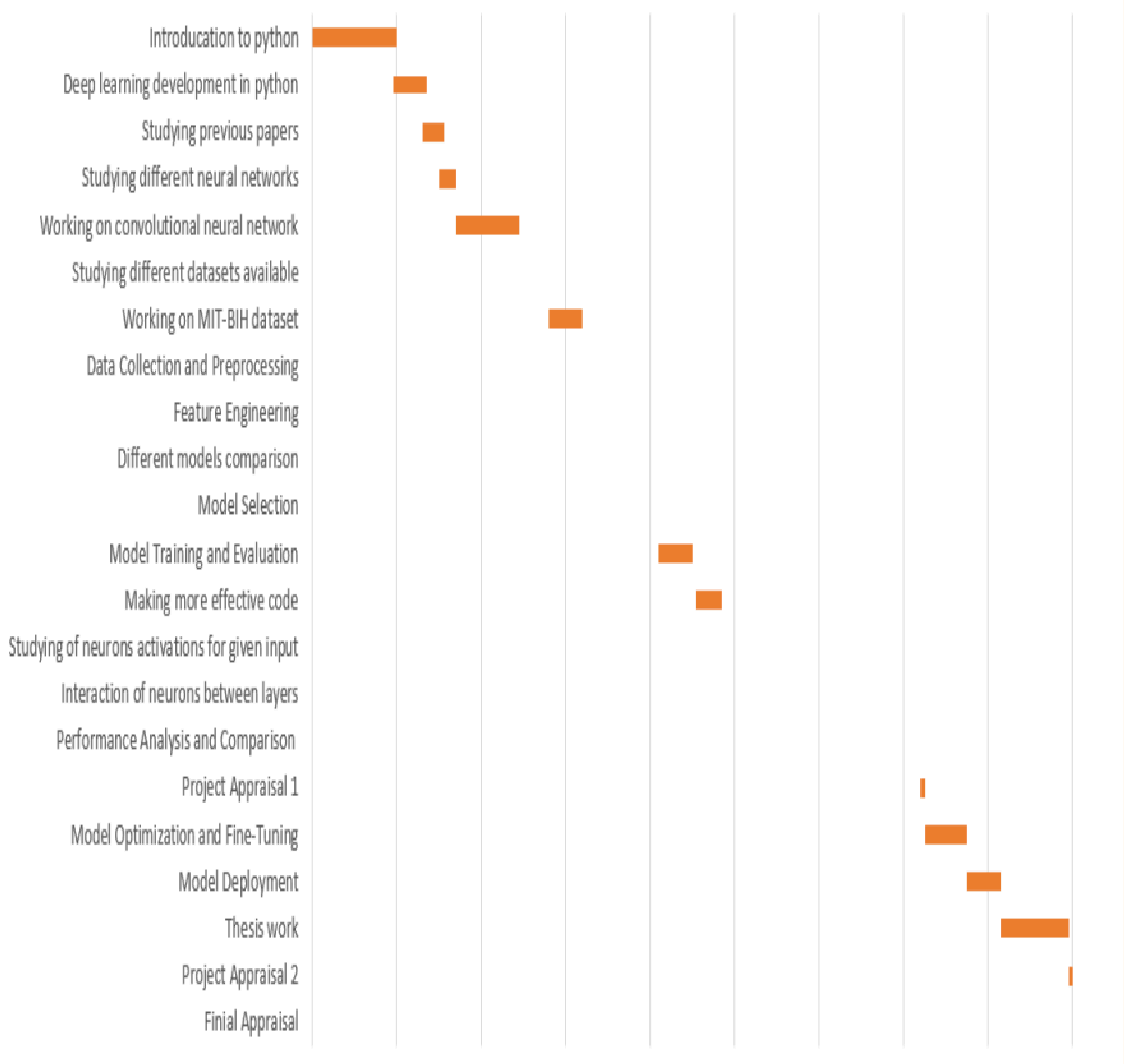
```
              precision    recall  f1-score   support

           0       0.95      0.97      0.96      1635
           1       0.81      0.92      0.86       780
           2       1.00      0.99      0.99      1538
           3       0.58      0.10      0.18       136
           4       0.00      0.00      0.00        26
           5       1.00      0.98      0.99       406

   micro avg       0.94      0.94      0.94      4521
   macro avg       0.72      0.66      0.66      4521
weighted avg       0.93      0.94      0.93      4521

Confusion matrix, without normalization
[[1582   37    0    6   10    0]
 [  51  718    6    4    1    0]
 [   2   14 1522    0    0    0]
 [  27   95    0   14    0    0]
 [   5   20    0    0    0    1]
 [   2    6    0    0    0  398]]
F1 score: [0.95762712 0.85988024 0.99282453 0.175      0.         0.98881988]
AUC for N: 0.9999909698392632
AUC for V: 0.9999880960883747
AUC for /: 1.0
AUC for A: 1.0
AUC for ~: 1.0
```

Deployment Using Flask

- In the image, the signals have 57 peaks, and the signal of each peak was estimated by the trained model.
- Then the whole signals consists of 27 normal beats, 6 venticular beats, 2 atrial beats, and 22 noises signals.
- If the signals represent noise, the most of the signals should have been estimated as noise.
- Then, the patient with the signals must be healthy or possibly have venticular contraction.



PROJECT ACTIVITIES TIMELINE



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Feedback Form

Thank You